

BOURG ST-MAURICE, France

WMO: 074970

Lat: 45.6126N

Lon: 6.7633E

Elev: 2848

StdP: 13.25

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.9	17.6	6.3	8.4	24.2	9.7	10.0	24.8	19.1	42.8	16.4	43.5	1.7	350	0.488

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.1	88.9	64.3	85.5	63.6	82.2	62.7	66.4	83.3	65.0	81.0	63.7	78.6	5.4	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.1	89.5	69.5	59.8	85.3	68.5	58.5	81.3	67.7	32.7	83.7	31.5	81.0	30.5	79.3	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.9	13.1	10.1	DB	9.2	93.6	5.9	3.6	4.9	96.2	1.4	98.3	-1.9	100.3	-6.2	102.9	(4)
(5)			WB	8.1	69.8	5.7	1.3	4.0	70.7	0.6	71.5	-2.6	72.2	-6.8	73.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.6	33.5	36.0	43.7	50.0	57.1	64.3	67.4	66.4	59.6	52.5	41.6	34.7		
	DBStd	13.66	6.18	7.03	6.72	6.64	6.40	6.62	6.03	5.96	5.94	6.13	7.56	6.62		
	HDD50	2006	512	394	213	80	14	1	0	0	5	47	265	475		
	HDD65	5554	976	813	662	449	255	92	45	54	180	388	701	939		
	CDD50	2239	0	0	16	81	234	428	540	509	291	124	15	1		
	CDD65	312	0	0	0	0	9	69	120	98	16	0	0	0		
	CDH74	3578	0	0	0	14	172	848	1284	1033	218	9	0	0		
	CDH80	1184	0	0	0	0	31	275	456	382	40	0	0	0		
(14) Wind	WSAvg	3.3	2.7	3.1	3.8	4.4	3.7	3.3	3.2	2.8	2.8	3.2	3.3	2.9		
(15) Precipitation	PrecAvg	49.0	4.2	3.8	3.5	3.5	4.5	4.4	4.3	3.9	3.7	3.9	4.6	4.7		
	PrecMax	62.2	11.6	14.1	11.0	6.3	8.7	7.8	7.3	7.2	7.7	9.3	12.0	14.2		
	PrecMin	37.5	0.5	0.6	0.8	0.4	1.7	2.5	1.6	1.1	1.3	0.7	1.3	0.0		
	PrecStd	6.2	2.9	3.0	2.6	1.6	1.9	1.5	1.4	1.4	1.6	2.1	2.7	3.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.9	60.4	69.6	75.9	84.0	91.9	93.3	93.3	84.7	74.9	64.4	55.0		
		MCWB	42.6	44.2	48.6	53.5	60.7	64.3	64.5	64.6	62.6	56.7	49.7	43.5		
	2%	DB	49.0	54.6	65.3	71.3	79.0	86.9	88.8	88.6	80.1	70.4	59.9	51.6		
		MCWB	40.0	41.9	47.1	51.6	58.2	64.0	64.3	64.1	61.2	55.4	47.5	41.7		
	5%	DB	45.4	50.2	60.9	67.4	74.3	82.9	85.1	84.2	76.0	66.8	56.8	47.3		
		MCWB	37.6	39.7	45.5	49.8	56.3	63.0	63.7	63.7	59.4	53.9	46.0	39.5		
	10%	DB	42.1	46.4	56.2	63.6	70.6	79.0	81.5	80.0	71.8	63.3	53.5	43.5		
		MCWB	36.3	37.6	43.3	48.1	54.4	61.2	62.8	62.4	57.9	52.3	44.6	38.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	45.8	50.4	54.8	62.6	67.9	68.3	68.2	64.7	59.5	51.6	45.1		
		MCDB	50.0	56.4	64.8	72.8	79.4	85.3	85.3	85.6	78.7	69.5	59.2	51.6		
	2%	WB	41.2	42.8	48.0	52.5	59.8	65.6	66.3	66.2	62.3	57.3	48.7	43.0		
		MCDB	46.5	52.2	61.6	69.0	75.0	83.1	83.4	82.3	76.9	66.2	56.8	48.9		
	5%	WB	39.1	40.7	46.3	50.7	57.6	63.8	64.8	64.7	60.6	55.4	47.0	41.1		
		MCDB	43.8	48.8	58.7	65.1	70.6	80.0	81.1	79.6	72.9	63.7	54.2	45.6		
	10%	WB	37.1	38.5	44.4	48.9	55.6	62.1	63.5	63.2	58.8	53.6	45.4	39.1		
		MCDB	41.0	45.0	54.3	61.5	67.8	76.4	78.5	76.6	69.1	61.1	52.2	42.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.0	17.6	21.0	20.9	21.6	23.7	24.1	23.5	22.0	20.1	15.6	13.5		
		MCDBR	17.7	24.2	28.8	28.2	29.0	30.2	30.1	30.9	28.7	25.1	20.5	17.5		
		MCWBR	10.9	13.6	13.7	12.2	11.8	12.0	10.8	11.2	12.0	12.4	10.7	10.4		
	5% WB	MCDBR	15.4	21.7	25.5	26.1	26.1	27.9	27.1	27.1	26.2	22.4	18.5	14.1		
		MCWBR	10.2	12.7	12.6	11.8	11.3	11.8	11.0	11.0	11.5	11.6	10.2	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.300	0.355	0.406	0.421	0.421	0.411	0.396	0.376	0.350	0.307	0.270		
	taud		2.298	2.212	2.120	2.101	2.148	2.208	2.243	2.293	2.337	2.379	2.400	2.353		
	Ebn at Noon		266	277	274	269	269	269	270	270	266	256	250	253		
	Edh at Noon		25	33	42	47	47	44	42	39	34	28	23	21		
(44) All-Sky Solar Radiation	RadAvg		513	786	1128	1400	1603	1916	1953	1702	1320	856	511	429		
	RadStd		59	98	143	171	135	149	145	113	102	98	76	56		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (73 neighbors)	N/A	N/A	N/A	+1.30	+0.61	+0.41	N/A	N/A	+108	+36

Nomenclature: See separate page